GravelKing Pro

MLK V3.5 — Signal-Level Audio Security

Technical Brief for Enterprise Partners

All N One LLC · GravelKing Productions · 2026

$43B	LSB	FRE 901(b)(9)	< 2ms
Creator Economy	Signal-Level	Court-Admissible	Verification

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SECTION 1

Executive Summary & Market Imperative

The 2026 AI content landscape has created an urgent trust deficit for digital creators. Generative AI tools now produce audio indistinguishable from human-authored recordings, AI training datasets ingest unlicensed catalog without attribution, and platform-level metadata (ID3 tags, ISRC codes) can be stripped, reassigned, or fraudulently claimed by any actor with basic tooling.

GravelKing Pro MLK V3.5 solves this at the signal level — not the metadata level. Rather than relying on external ledgers, blockchain timestamps, or third-party registries that can be circumvented, MLK V3.5 embeds cryptographic proof of ownership directly into the audio bitstream using dual-anchor LSB steganography. This proof travels with the track permanently, survives platform distribution, and is verified server-authoritatively — making chain-of-custody disputes resolvable without court-ordered discovery.

Market Context

- $43B+ global creator economy (2026), growing 18% YoY
- EU AI Act (enforcement Aug 2026) requires provenance declarations for AI-assisted content
- U.S. Copyright Office: Thaler v. Vidal (2023) establishes human-authorship threshold requirement
- Platform DSPs face regulatory pressure to verify content provenance before monetization
- PortalBunny and similar creator ecosystems require native authenticity infrastructure as they scale

SECTION 2

Ingestion Defense & Bad-Actor Mitigation

Before any signal-level watermark is embedded, every asset passes through a three-tier ingestion validation pipeline. This prevents adversarial actors from registering stolen, AI-generated, or commercially licensed content and receiving a fraudulent GravelKing certificate.

Tier 1 — Metadata Collision Check

Scans all submitted metadata fields (filename, title, artist, ISRC, ISWC, UPC) against a curated registry of known commercial patterns. Detects:

- ISRC format collisions (format: CC-XXX-YY-NNNNN)
- Major label/DSP identifiers (Sony, Universal, Warner, Atlantic, Columbia, etc.)
- Known artist name signatures in the commercial blacklist

Tier 2 — Spectral Audio Fingerprinting

Runs ffmpeg ebur128 loudness analysis on the submitted audio. Extracts integrated LUFS and true peak dBFS from the EBU R128 report. Rejects:

- Near-silence signals (integrated LUFS < -55) — copied carrier tracks or empty submissions
- Artificially maxed signals (peak dBFS > -0.5) — ripped/clipped commercial masters
- Single-frequency test tones — bot-generated submissions

Tier 3 — Digital Assertion Signature

Requires an explicit machine-readable copyright ownership assertion (author_assertion: true) in every ingest request. This field is included in the forensic certificate as a signed warrant. Absent the warrant, the server returns HTTP 422 ERR_COPYRIGHT_WARRANT_REQUIRED before any processing occurs.

SECTION 3

Court-Admissible Forensic Chain-of-Title

Dual-Anchor Split-Key Architecture

MLK V3.5 uses a nominator/denominator split: the two halves of the cryptographic proof are stored separately, making verification server-authoritative by design.

- Anchor A (Nominator) — embedded into the audio's least-significant bits. Encodes SHA-256(contentHash | artistHandle | certId), truncated to 32 hex chars. Survives lossless export (WAV, AIFF, FLAC). Travels with the signal, not the container.
- Anchor B (Denominator) — stored exclusively on GravelKing servers. Never transmitted to any client. The server-side complement required to verify the full handshake.
- HMAC-SHA256 Handshake — HMAC(SESSION_SECRET, certId | nominator | denominator). Tamper-evident binding between the in-signal anchor and the server record.

Verification Protocol

POST /api/kernel/verify-signal accepts any WAV file and returns a structured VerifyResult. Warrant status is one of:

- INTACT — Both anchors present, HMAC matches, content hash reconstructs correctly
- TAMPERED — Anchor A present, but Anchor B missing, HMAC mismatches, or content hash drift detected. Triggers Default of Warrant in downstream legal proceedings.
- NO_WATERMARK — No GravelKing LSB signature detected. Track was not issued by this platform.

Legal Standards Met

- FRE Rule 901(b)(9) — SHA-256 content fingerprint satisfies federal digital evidence authentication
- Thaler v. Vidal (2023) — Human authorship score >= 25% threshold for copyright assertability
- Dual-storage (PostgreSQL + Google Firestore) prevents spoliation objections
- ISO-8601 UTC certifiedAt timestamp predates any third-party claim

SECTION 4

B2B Platform Integration Model

GravelKing Pro is designed for direct API integration by creator economy platforms, DSPs, and brand-content marketplaces. The integration surface is a RESTful JSON API with multipart/form-data ingest.

Core Endpoints

- POST /api/v1/ingest (alias: /api/kernel/master) — Submit audio for MLK V3.5 processing + cert embedding. Returns processed stream + X-GK-CertId header.
- POST /api/v1/verify (alias: /api/kernel/verify-signal) — Public clean-room verification. No auth required. < 2ms median response for pre-certified tracks.
- GET /api/court-cert/:certId — Structured forensic certificate JSON.
- GET /api/court-cert/:certId.pdf — Court-ready PDF certificate.
- GET /api/v1/download-whitepaper — Stream this technical brief as a PDF.
- POST /api/v1/lead-capture — Enterprise onboarding. Issues 7-day demo API key.
- GET /api/v1/whitepaper-spec — Structured JSON data model of this document.

Supported Media Formats

- Lossless: WAV (PCM 16/24/32-bit), FLAC, AIFF — LSB embedding fully preserved
- Lossy (ingest-only): MP3, AAC, OGG, Opus, M4A — auto-converted to WAV pre-embedding
- Video containers (audio extraction): MP4, MOV, MKV, WebM — ffmpeg audio strip on arrival
- Sample rates: 44.1 kHz, 48 kHz, 96 kHz — all supported via ffmpeg normalization

Integration Flow for Creator Platforms

1. Platform calls POST /api/v1/ingest with creator's audio file + author_assertion: true
2. MLK V3.5 validates through the 3-tier ingestion filter
3. If cleared: applies mastering chain, embeds Anchor A watermark, stores Anchor B + cert
4. Returns processed audio to platform + certId header
5. Platform stores certId against the creator's content record
6. Any downstream dispute: call POST /api/v1/verify with the distributed file — warrant status returns in < 2ms

Throughput & SLA

- Median verification latency: < 2ms (pre-certified tracks, in-region)
- Mastering + embedding: 8-45 seconds per track depending on duration and format

- Concurrency: horizontal Docker scaling, up to 12 parallel processing lanes per node
- Webhook delivery for async cert events: configurable per API key

SECTION 5

Containerized Infrastructure Moat

The MLK V3.5 kernel is implemented as a native C-bound DSP pipeline exposed through a Node.js FFI layer. The architecture is designed for zero-trust enterprise cloud deployment and on-premise air-gapped installations.

Core Architecture

- Docker multi-stage build: Node.js 22 LTS + ffmpeg-headless + native kernel binary
- mmap() for zero-copy audio buffer access — no heap allocation for signal processing
- mlockall(MCL_CURRENT | MCL_FUTURE) — pins kernel pages, prevents swap exposure of cert material
- Process isolation: each ingest request runs in a sandboxed child process with resource limits
- Zero secrets in Docker image — all credentials injected at runtime

Deployment Targets

- SaaS (gravelkingpro.it.com) — immediate access, usage-based billing
- AWS ECS / GCP Cloud Run — container registry publish, auto-scale on CPU, 99.9% SLA
- On-premise Docker Compose — air-gapped for major labels and studios, no data egress
- White-label API — custom subdomain, volume pricing, dedicated webhooks, custom branding

Security Posture

- SESSION_SECRET-derived HMAC — never stored in application code or Docker image
- Database dual-write: PostgreSQL (queryable) + Google Firestore (append-only audit trail)
- Admin endpoints require separate ADMIN_KEY header — never exposed on public routes
- No telemetry to third parties — all processing is on-instance

Ready to integrate?

Start with the public verification demo, then request enterprise API access.

Demo Lab	gravelkingpro.it.com/verify
Enterprise Contact	kevm@gravelkingpro.it.com
Technical Brief	gravelkingpro.it.com/whitepaper
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